BanglaWheel

Platform Plan & Developer Specification — Version 1.1

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1. Platform Overview

BanglaWheel is a used car marketplace for Bangladesh built around one thing: buyer trust. The country has listing sites already — GarirBazar, Bhalogari, Car Haat, Bikroy Motors — but none verify sellers, inspect cars, or back listings with any guarantee. That gap is our entry point.

The platform combines professional car photography, NID-based seller verification, independent vehicle inspection, insurance-backed warranty through Mercantile Islami Insurance, live auction for dealer dead stock, and a personalised recommendation engine based on location and budget. The full platform is available in English and Bangla — switchable from the top navigation bar.

Three types of users:

- Buyer — browses, saves, bids, books inspections, files warranty claims, applies for financing.
- Seller / Dealer — submits listings, uploads media, pays warranty deposit, manages listings.
- Admin — approves listings, reviews KYC, assigns inspections, runs auctions, views analytics.

Design rule: mobile-first. 95%+ users are on Android. Every page must work on 360–414px. Target: under 2 seconds load on a mid-range phone.

2. Tech Stack

Layer	Tool	Reason
Frontend	Next.js 14 (App Router)	SSR — every listing indexed by Google
Mobile	PWA	Installable on Android, no Play Store
Backend	Node.js + Express	APIs and real-time auction events
Database	PostgreSQL + Redis	Main store + Redis for live bid caching
File Storage	AWS S3 / Cloudflare R2	Photos, videos, KYC documents
Search	Meilisearch	Fast, typo-tolerant, Bengali+English queries
Payments	SSLCommerz	bKash, Nagad, Rocket, Visa, Mastercard
KYC	KYC.bd API	Real-time NID check vs Election Commission DB
Maps	Google Maps API	Geocoding addresses, distance calculation
Auth	JWT + SMS OTP	Phone OTP — most reliable auth in Bangladesh
Real-time	Socket.io	Live bid updates to all open browsers
CDN / Security	Cloudflare	SSL, caching, DDoS protection
i18n	next-intl	Language toggle EN/Bangla — see Section 3
Hosting	DigitalOcean (start)	Migrate to AWS when traffic grows
Blog / CMS	Sanity.io	Admin writes articles without touching code

3. Language Toggle — English / ■■■■■

Every page on BanglaWheel is available in both English and Bangla. The user switches language using a toggle in the top navigation bar — instant switch, no page reload. The choice is saved to their session and account.

Where the toggle lives

Top right of the navigation bar, before the login button. A simple two-state pill:

When Bangla is active, the Bangla pill turns navy and EN becomes grey. One tap to switch back.

Library to use: next-intl

- npm install next-intl
- Set up routing.ts with locales: ["en", "bn"]
- Create /messages/en.json — all text keys in English. This is the master file.
- Create /messages/bn.json — translate all keys to Bangla. Use a native speaker to review.
- Replace every hardcoded string in every component with t("key") calls.
- On toggle click: setLocale("bn") or setLocale("en"). Save to localStorage and user account.
- Load Hind Siliguri via next/font. Apply class bn-font to html element when locale is "bn".
- Add hreflang meta tags for EN and Bangla — required for Google to index both language versions.

What needs to be translated

Section	What to Translate
Navigation & menus	All nav links, dropdowns, footer links
Homepage	Hero headline, search placeholder, section headings, CTA buttons, trust stats
Browse page	Filter labels, sort options, result count, card labels
Car detail page	Section headings, badge text, button labels, specification table keys
Listing form	Step titles, field labels, dropdown options, help text, error messages
Auction page	Timer labels, bid button, bid history, status messages
Account dashboard	Tab labels, status labels, notification text
Education Hub	Article titles and body in both languages via CMS — admin writes two versions per article.
Error messages	Form validation, 404 page, loading states, success confirmations
Legal pages	Privacy Policy and Terms of Service must both have a Bangla version.

Bangla font: use Hind Siliguri or Noto Sans Bengali (both Google Fonts, free, render cleanly on Android). Load via next/font.

Admin panel stays English only in V1. Add Bangla admin panel in V2 if needed.

4. All Pages & Routes

Route	Access	Purpose
/	Public	Homepage — search, featured listings, trust section
/cars	Public	Browse all listings with full filter panel
/cars/[slug]	Public	Single car detail page
/sell	Registered	Multi-step listing submission form
/dealers	Public	Directory of all verified dealers
/dealers/[slug]	Public	Individual dealer page — listings, reviews
/auction	Public	All active and upcoming auctions
/auction/[id]	Public	Live auction page with real-time bidding
/certified	Public	What BanglaWheel Certified means
/garages	Public	Certified Garage Partner directory with map
/inspect	Registered	Book an inspection
/warranty	Public	Warranty and guarantee program
/financing	Public	Financing partners + EMI calculator
/paperwork	Public	BRTA document guide, ownership transfer steps
/learn	Public	Buyer Education Hub — article index
/learn/[slug]	Public	Individual guide article (CMS)
/compare	Registered	Side-by-side comparison — up to 3 cars
/account	Registered	User dashboard home
/account/listings	Seller	Active listings, status, analytics
/account/bids	Buyer	Active bids and bid history
/account/warranty	Buyer	Active warranties, claim submission, status
/account/documents	Seller	KYC documents and verification status
/account/settings	Registered	Profile, notifications, language preference, budget
/admin	Admin	Admin dashboard
/admin/listings	Admin	Listing moderation queue
/admin/sellers	Admin	Seller KYC review
/admin/inspections	Admin	Inspection job assignment
/admin/auctions	Admin	Auction control panel

Route	Access	Purpose
/admin/analytics	Admin	Traffic and conversion analytics
/privacy-policy	Public	Required legal page (EN + Bangla)
/terms-of-service	Public	Required legal page (EN + Bangla)
/data-deletion	Public	Data deletion request form
/about	Public	Company info, address, trade licence

5. Listing Submission Form

Eight-step flow with progress bar. All spec fields are structured dropdowns. Keeps data consistent for search and the recommendation engine. Full form available in EN and Bangla.

Step 1 — Vehicle Identity

Field	Type	Options
Make / Brand	Dropdown	Toyota · Honda · Nissan · Mitsubishi · Suzuki · BMW · Mercedes · Audi · Hyundai · Kia · Mazda · Subaru · Lexus · VW · Ford · Other
Model	Dropdown (dynamic)	Loads based on Make
Year	Dropdown	1990 – 2026
Variant / Grade	Text	e.g. Allion A15 G Package
Chassis / VIN	Text	Required for BRTA cross-check.
Registration No.	Text	e.g. Dhaka Metro-Ga-11-1234
Auction Grade	Dropdown	6 · 5 · 4.5 · 4 · 3.5 · 3 · 2 · 1 · R · RA · N/A
Country of Origin	Dropdown	Japan · South Korea · Germany · UK · India · Other
Import Type	Dropdown	Reconditioned (JDM) · Brand New · Local second-hand

Step 2 — Engine & Drivetrain

Field	Type	Options
Displacement	Dropdown	Under 660cc / 660–1000 / 1001–1300 / 1301–1500 / 1501–1800 / 1801–2000 / 2001–2500 / Over 2500
Fuel Type	Dropdown	Petrol · Diesel · Hybrid · CNG · Electric · Petrol+CNG
Transmission	Dropdown	Automatic · Manual · CVT · DCT
Drive Type	Dropdown	FWD · RWD · AWD · 4WD
Engine Code	Text	Optional — e.g. 1NZ-FE
Cylinders	Dropdown	3 · 4 · 6 · 8 · Other
Horsepower	Number	In bhp
Turbo	Toggle	Yes / No

Step 3 — Condition & History

Field	Type	Options
Odometer Reading	Number	In kilometres

Field	Type	Options
Odometer Verified	Toggle	Yes / Not verified
Previous Owners	Dropdown	1 · 2 · 3 · 4+
Accident History	Dropdown	No accident · Minor (repaired) · Major (repaired) · Unknown
Flood History	Dropdown	No · Yes · Unknown
Overall Condition	Dropdown	Excellent · Very Good · Good · Fair · Needs Work
Last Service Date	Date	
Service Records	Toggle	Available / Not available

Step 4 — Body & Exterior

Field	Type	Options
Body Type	Dropdown	Sedan · Hatchback · SUV · MPV · Pickup · Wagon · Coupe
Exterior Colour	Dropdown + swatch	White · Silver · Black · Grey · Red · Blue · Gold · Other
Interior Colour	Dropdown	Beige · Black · Grey · Brown · Other
Upholstery	Dropdown	Fabric · Leather · Half-Leather · Synthetic Leather
Doors	Dropdown	2 · 4 · 5
Seating Capacity	Dropdown	2 · 4 · 5 · 6 · 7 · 8

Step 5 — Features Checklist

Grouped checkboxes. Stored as JSON. Shown as chips on the detail page.

Category	Checkboxes
Safety	ABS · Airbags · ESC · Reverse Camera · Parking Sensors · Lane Departure · Blind Spot · Collision Warning · Hill Start
Comfort	Auto AC · Heated Seats · Power Seats · Sunroof · Push-Button Start · Keyless Entry
Technology	Navigation · Touchscreen · CarPlay · Android Auto · Bluetooth · USB · Digital Dashboard
Exterior	Alloy Wheels · LED Headlights · Fog Lights · Roof Rails · Tinted Windows
Other	CNG Kit · Spare Tyre · Owner's Manual

Step 6 — Price & Location

Field	Type	Notes
Asking Price	Number (BDT)	Raw number only
Price Negotiable	Toggle	Yes / No

Field	Type	Notes
Division	Dropdown	Dhaka · Chittagong · Sylhet · Rajshahi · Khulna · Barisal · Rangpur · Mymensingh
District	Dropdown	Loads based on Division
Area / Thana	Text	Geocoded via Google Maps API — lat/lng stored
Preferred Contact	Dropdown	WhatsApp only · Phone only · Both
Phone Number	Input	Visible only to logged-in users who click "Show Number"

Step 7 — Media Upload

- Minimum 8 photos. Maximum 30. Required slots: Front · Rear · Driver Side · Passenger Side · Front Interior · Rear Interior · Dashboard · Odometer · Engine Bay · Trunk · Additional.
- Video: 1 file, MP4 only, max 100MB.
- Admin rejects blurry, dark, or missing-angle photos.
- "Request BanglaWheel Professional Shoot" toggle — listing stays Draft until shoot is done.

Step 8 — Seller Declaration

- "I confirm this vehicle is legally mine to sell."
- "All information provided is accurate to the best of my knowledge."
- "I agree to BanglaWheel's Terms of Service and Privacy Policy."
- "I consent to my NID being verified via Bangladesh Election Commission database."

6. Car Detail Page

Display order — top to bottom.

Component	What It Does
Hero Gallery	Full-width swipeable photos. Lightbox on desktop, swipe on mobile. Video inline. Share (WhatsApp, Facebook, Copy Link). Save/Wishlist heart (login required).
Quick Stats Bar	Sticky: Price · Year · KM · Fuel · Transmission · Location.
Trust Badges	■ BanglaWheel Certified · Verified Seller · Warranty Covered · Premium · BanglaWheel Bid. Only show applicable badges.
Credibility Report	Expandable. Inspection date · Garage name · Section scores · Overall /100. Download PDF button.
Full Specification	Two-column table: Vehicle Identity · Engine · Body · Features.
Warranty Section	If covered: period, what is covered (Engine, Transmission, Electrical, AC) and what is not (tyres, brakes, buyer damage). Mercantile Islami logo.
Paperwork Checklist	Collapsed by default. Lists BRTA documents buyer should receive.
Seller Panel	Name or dealer logo · Verified badge · Member since · Total listings · WhatsApp/Phone (login-gated) · Report listing.
Similar Cars	6 listings: same make/model, similar price, same city. Powered by recommendation engine.
Enquiry / Contact	Send message form · WhatsApp button · "Book inspection" → /inspect pre-filled.

7. Search, Filters & Sort

- Prominent search bar on homepage and in top nav on all pages.
- Autocomplete: brand names, model names, popular phrases as user types.
- Suggested pills below bar: "Toyota Allion Dhaka" · "Hybrid under Tk 12 lakh".
- Search bar placeholder switches with language toggle.
- All queries logged — feeds content and inventory strategy.

Filter	UI	Options
Make	Multi-select	All brands
Model	Multi-select	Dynamic, loads by Make
Year Range	Dual slider	1995 – 2026
Price Range	Dual slider	BDT 50,000 – 1 crore+
Mileage Range	Dual slider	0 – 300,000 km
Fuel Type	Checkboxes	Petrol · Diesel · Hybrid · CNG · Electric
Transmission	Checkboxes	Automatic · Manual · CVT
Body Type	Icon selector	Sedan / SUV / Hatchback / MPV / Pickup
Condition	Checkboxes	Excellent · Very Good · Good
Location	Division → District	District loads dynamically
Distance	Slider	10 · 25 · 50 · 100 km · Anywhere
Listing Type	Checkboxes	Certified · Verified Seller · BanglaWheel Bid
Auction Only	Toggle	Live auctions only
Warranty Available	Toggle	Warranty-covered cars only

Sort: Newest Listed (default) · Price Low–High · Price High–Low · Mileage Lowest · Most Viewed · Auction Ending Soonest · Distance Nearest First.

Results: 3-col grid desktop / 1-col mobile. Toggle to list view. Infinite scroll. Active filters as removable chips.

8. Smart Recommendation Algorithm

Two signals combined into one score. Track A = where the user is. Track B = what they can afford and like.

Track A — Location & Distance

Browser requests GPS on load. Falls back to registered Division or IP geolocation if denied. Every listing stores lat/lng geocoded from the seller's area. Distance via Haversine formula as a PostgreSQL function.

Haversine Function (PostgreSQL)

```
CREATE OR REPLACE FUNCTION haversine_km(
  lat1 FLOAT, lng1 FLOAT, lat2 FLOAT, lng2 FLOAT
) RETURNS FLOAT AS $$
DECLARE
  R    FLOAT := 6371;
  dlat FLOAT := radians(lat2 - lat1);
  dlng FLOAT := radians(lng2 - lng1);
  a    FLOAT;
BEGIN
  a := sin(dlat/2)^2
     + cos(radians(lat1)) * cos(radians(lat2)) * sin(dlng/2)^2;
  RETURN R * 2 * atan2(sqrt(a), sqrt(1-a));
END; $$ LANGUAGE plpgsql IMMUTABLE;
```

- Default radius: 25 km. Adjustable via Distance filter.
- "Cars Near You" on homepage: 6 cards, closest first.
- Distance badge on each card: "8 km away."
- If location unknown: fall back to registered Division, sort by freshness.

Track B — Affordability & Preference

Builds a silent budget profile from behaviour. No declaration needed, though users can set one in account settings.

Signal	Weight	How Measured
Price range of listings viewed	70%	Average price of last 10 listings where user spent 30+ seconds
Search query text	15%	Parses keywords: "under 10 lakh", "budget car"
Saved/wishlisted cars	10%	Price of explicitly saved cars
Explicit budget (account settings)	5%	Optional field

Affordability band: Midpoint x 0.6 to Midpoint x 1.4. Cars outside that range get a reduced score.

Preference matching:

- Make — Toyota viewed 3x → Toyota boosted in results.
- Body type — sedan vs SUV vs hatchback click patterns.
- Fuel type — hybrid vs petrol vs CNG interest.
- Feature matching — user saves cars with Navigation → listings with Navigation rank higher.
- Condition floor — user views Excellent/Very Good only → Fair listings suppressed.

Final Score Formula

```
Final Score = (Location x 0.35) + (Affordability x 0.40) + (Preference x 0.25) Location Score = 1 / (1 + distance_km) Affordability = 1.0 inside band / 0.5 just outside / 0.0 far beyond Preference = matched signals / total possible signals
```

Where recommendations appear:

- "Recommended For You" — Homepage: 6 personalised cards.
- "Cars Near You" — Homepage: 6 cards by distance only.
- "Similar Cars" — Detail page: same make/model, similar price, same city.
- SMS/push alert (opt-in): "New Toyota Allion listed 12 km from you within your budget."

Phase	What to Build
Phase 1	Rule-based. Division = location. Last search price = budget. Make/model from query = preference.
Phase 2	Behavioural tracking. Log listing_viewed, listing_saved, enquiry_sent. Nightly affordability midpoint.
Phase 3	Collaborative filtering. Requires 5,000+ interactions. Use Recombee or scikit-learn.

9. Verified Seller & KYC

Verification is optional but incentivised — verified sellers get a shield badge and rank higher.

#	Step	Detail
1	Phone OTP	Enter mobile → 6-digit OTP → verified.
2	NID Input	NID number, full name as on NID, date of birth.
3	API Check	KYC.bd API verifies against Bangladesh Election Commission.
4	Document Upload	Front of NID + Back + Selfie holding NID.
5	Admin Review	Checks selfie against NID photo. Approves or rejects.
6	Badge Granted	"Verified Seller" shield on profile and all listings.

Dealer KYC — extra documents needed:

- Trade Licence (City Corporation or Municipality)
- TIN Certificate
- Owner's NID (same individual flow)
- Company letterhead scan
- Physical address confirmed via Google Maps pin
- On approval: Dealer Profile Page created automatically

NID numbers encrypted AES-256. Documents in private S3 bucket. Every access logged.

10. 100-Point Inspection & Credibility Report

Inspections done by Certified Garage Partners. Garage fills all 100 points on a mobile web form and uploads issue photos. Admin reviews before publishing.

- Seller requests inspection. Listing stays in Draft.
- Admin assigns nearest Certified Garage.
- Seller brings car to garage.
- Inspector fills 100-point form on BanglaWheel Garage Portal. Uploads photos.
- Admin reviews and approves publication.
- "Certified" badge if score is 70/100 or above.

Scoring: Pass = 1 pt · Advisory = 0.5 pts · Fail = 0 pts

Score	Status
90–100	Excellent — BanglaWheel Certified
80–89	Very Good — BanglaWheel Certified
70–79	Good — Certified with Advisories
60–69	Fair — Inspected, Not Certified
Below 60	Significant Issues — Not Certified

Section A — Underhood (20 points)

- 1. Battery condition and terminal corrosion
- 2. Alternator operation
- 3. Engine oil level and condition (colour, no froth, no metal flakes)
- 4. Oil cap — frothy residue indicates blown head gasket
- 5. Coolant level and condition
- 6. Coolant hoses — no cracks, no bulges
- 7. Radiator condition and fan operation
- 8. Power steering fluid
- 9. Brake master cylinder fluid
- 10. Air filter condition
- 11. Timing belt/chain condition
- 12. Engine mount condition
- 13. Transmission mount condition
- 14. Fuel system — no leaks
- 15. AC compressor operation
- 16. AC condenser condition
- 17. Engine noise on cold start (ticking, grinding, knocking)
- 18. Engine noise on warm start

- 19. Idle stability
- 20. Exhaust smoke: blue=oil, white=coolant, black=rich fuel

Section B — Under Vehicle (15 points)

- 1. Engine oil leaks from below
- 2. Transmission fluid leaks
- 3. Coolant leaks
- 4. Differential fluid leaks (AWD/4WD)
- 5. Exhaust — pipe, muffler, hangers
- 6. Frame/unibody — no kinks from collision
- 7. Front strut/shock absorber
- 8. Front ball joint and tie rod
- 9. Front suspension bushings
- 10. Rear suspension
- 11. CV joint boots and axle
- 12. Brake calipers front
- 13. Brake calipers rear
- 14. Brake rotors front
- 15. Brake rotors rear

Section C — Tyres & Wheels (10 points)

- 1. Front left tyre tread depth (min 3mm)
- 2. Front right tyre tread
- 3. Rear left tyre tread
- 4. Rear right tyre tread
- 5. All 4 tyres same brand and size
- 6. Tyre sidewall — no bulges or cuts
- 7. Wheel/rim — no cracks, no bends
- 8. Lug nut tightness
- 9. Spare tyre present and inflated
- 10. Valve stem condition

Section D — Body & Paint (15 points)

- 1. Panel gaps — bumper to bonnet
- 2. Panel gaps — doors
- 3. Panel gaps — rear
- 4. Paint consistency — repainted panels show different shade
- 5. Bondo/filler — magnet test all panels
- 6. Windshield — chips, cracks
- 7. Front bumper
- 8. Rear bumper

- 9. Bonnet/hood alignment
- 10. Driver door
- 11. Passenger door
- 12. Rear doors
- 13. Boot lid and seal
- 14. Roof — dents, rust, water channel
- 15. Headlights — no cracks, no fogging

Section E — Lights & Electricals (10 points)

- 1. High beam and low beam both sides
- 2. Front turn signals
- 3. Front fog lights (if equipped)
- 4. Brake lights including third brake light
- 5. Reverse lights
- 6. Rear turn signals
- 7. Parking/running lights
- 8. Dashboard warning lights (none stay on after start)
- 9. Interior dome and map lights
- 10. Hazard lights

Section F — Interior (15 points)

- 1. Water ingress — under carpets, seats, boot carpet (flood detection)
- 2. Front seat upholstery — cuts, stains, burns
- 3. Rear seat condition
- 4. Carpet front and rear
- 5. Headliner — no sagging, no stains
- 6. Dashboard — no cracks
- 7. Steering wheel wear
- 8. Power windows all doors
- 9. Central locking all doors
- 10. AC cooling — vent below 10 degrees C at max
- 11. Heater operation
- 12. Audio system / head unit
- 13. Sunroof open/close/tilt
- 14. Rear defrost
- 15. All gauges

Section G — Road Test (15 points)

- 1. Cold start — immediate, no excessive cranking
- 2. Transmission shift quality — smooth, no slipping
- 3. Clutch (manual only)

- 4. Acceleration — no hesitation under load
- 5. Steering effort and play
- 6. Steering return-to-centre
- 7. Regular braking — smooth, no pull
- 8. Hard braking — ABS activation (if equipped)
- 9. Suspension over bumps — no thuds
- 10. Highway noise
- 11. Full-lock turning both directions
- 12. Parking brake holds on incline
- 13. OBD-II scan — no active fault codes
- 14. Post-drive engine bay — no new leaks
- 15. Cruise control (if equipped)

11. BanglaWheel Bid — Live Auction

Auctions clear dealer dead inventory — cars unsold 45–60+ days. Anyone can watch live without registering.

#	Step	Detail
1	Seller submits	Normal listing or admin flags dead inventory (60+ days).
2	Auction setup	Starting Price · Reserve Price (hidden) · Duration (24h / 48h / 72h / 7 days).
3	Admin approves	Same KYC + listing review. Admin schedules start.
4	Goes live	Timer starts. Badge on listing. Shown on /auction page.
5	Bidding	Real-time via Socket.io. Auto-bid available.
6	Ends	Highest bidder notified. 24h to pay deposit via SSLCommerz.
7	No deposit	Winner misses 24h → second-highest offered car.
8	Confirmed	Admin marks listing Sold.

- Countdown timer — HH:MM:SS, every second.
- Current highest bid — updates instantly via Socket.io.
- Bid history — anonymised: "Buyer_DK_**82 — Tk 8,50,000 — 14 min ago."
- Auto-bid — user sets max, system bids minimum increments up to that.
- Anti-sniping: bid in final 5 min → timer extends 10 min.
- Seller cannot bid on their own listing. One account per phone number.

12. Warranty, Escrow & Mercantile Islami Insurance

Seller pays a refundable deposit at listing. Held in BanglaWheel escrow. No claims = deposit returned. Valid claims paid from deposit. Excess covered by Mercantile Islami Insurance.

Car Price	Deposit	Coverage
Under Tk 5 lakh	Tk 10,000	90 days
Tk 5–10 lakh	Tk 20,000	90 days (Phase 2: 6 months)
Tk 10–20 lakh	Tk 35,000	6 months
Above Tk 20 lakh	Tk 50,000	6 months

Covered: Engine failure · Transmission failure · Major electrical failure · AC compressor.

Not covered: Tyres, brakes, filters · Buyer-caused damage · Post-purchase accidents.

Claim flow: Buyer opens claim → Admin assigns Certified Garage → Estimate → Approved → Repair → Cost from escrow. Status: Submitted → Under Review → Approved → Repair Scheduled → Resolved.

13. Buyer Education Hub & BRTA Paperwork Guide

Articles via Sanity.io CMS. Admin writes in EN and Bangla. Google Article structured data on every article.

Article	SEO Target
"The Complete Used Car Buying Guide Bangladesh 2026"	Targets "used car buying bangladesh"
"Toyota Allion vs Honda Fit: Best Used Cars in Bangladesh"	High search volume
"What Does Auction Grade Mean? Japan Grades Explained"	Targets "auction grade"
"How to Transfer Car Ownership at BRTA"	Targets "BRTA ownership transfer"
"10 Red Flags When Buying a Used Car in Dhaka"	Targets "used car scam bangladesh"
"How to Spot a Flood-Damaged Car"	Post-monsoon relevance
"Used Car Prices Bangladesh — Monthly Market Report"	Repeat traffic

BRTA Document	What to Check
Registration Certificate (both copies)	Chassis must match. Not flagged or mortgaged.
Tax Token (current year)	Chassis matches. Not expired.
Fitness Certificate	Not expired.
Insurance Certificate	Valid, chassis matches.
Form T.T.O.	Seller signs — Transfer of Title Order.
Seller's NID copy	Name and photo match the person selling.
Non-Judicial Stamp Affidavit	Sworn statement of sale. Required by BRTA.
Bank NOC (if under loan)	Lender issues No-Objection Certificate.

14. Financing Partners & EMI Calculator

Bank	Product	Max Loan	Tenure	Type
IFIC Bank	HPSM Auto Finance	Up to Tk 60 lakh	5 years	Islamic
IPDC Finance	Vehicle Finance	Up to Tk 50 lakh	5 years	Conventional
BRAC Bank	Auto Loan	Up to Tk 40 lakh	4 years	Conventional
Dutch-Bangla Bank	Auto Loan	Variable	4 years	Conventional

- "Apply Now" on each bank — name, phone, target car → sent as lead.
- EMI Calculator: loan amount + down payment + tenure → monthly instalment.
- Required disclaimer: "Actual rates subject to bank approval."

15. User Accounts & Dashboards

Feature	Detail
Registration	Mobile OTP (required) · Email (optional) · Facebook · Google
Buyer Dashboard	Saved cars · Active bids · Warranties · Enquiry history · Language preference
Seller Dashboard	Listings with status · Edit listing · Enquiries · Analytics · Warranty deposit status
Notification Triggers	Listing approved/rejected · New enquiry · Bid placed · Outbid · Warranty update · KYC approved
Notification Channels	In-app bell · SMS · Email (optional)

16. Admin Panel

Module	Detail
Overview	Live stats: listings · pending · registrations · auctions · claims · escrow total.
Listing Management	Filter by status/date/location. One-click approve/reject. Edit any field.
Seller Management	KYC status. View documents. Approve/reject. Ban account.
Inspection Queue	Requested → Assigned → In Progress → Submitted → Published.
Auction Control	All active auctions. End early. Declare winner. Reserve Not Met handling.
Analytics	Top viewed listings · Top queries · Registrations · Funnel.
Garage Portal	Mobile login for Certified Garages. Fills 100-pt form. Cannot edit after submit.

17. SEO & Legal Pages

- Every listing meta title: "[Year] [Make] [Model] for sale in [City] — BanglaWheel"
- Meta descriptions auto-generated from listing fields.
- XML sitemap auto-generated and updated daily.
- Google Vehicle Listing structured data (JSON-LD) on every car page.
- Open Graph + Twitter Card — car photo becomes share preview on Facebook/WhatsApp.
- hreflang meta tags for EN and Bangla versions — required for Google indexing both languages.
- Privacy Policy and Terms of Service available in English and Bangla.
- Cookie consent banner on first visit.
- Footer: Trade Licence · TIN · Company registration · KYC.bd logo · Mercantile Islami logo.

18. Build Phases — V1 / V2 / V3

V1.0 — Launch (Weeks 1–10)

- Homepage, browse page, car detail page
- Listing submission form (all 8 steps)
- User registration and login (mobile OTP)
- Seller KYC flow
- Dealer profile pages
- Admin panel — listing approval and KYC review
- SSLCommerz payments
- WhatsApp enquiry on every listing
- Language toggle EN / Bangla (next-intl) — included from day one
- Hind Siliguri font for Bangla mode
- Full translation in /messages/bn.json
- Mobile-responsive on all screen sizes
- SEO: meta tags + sitemap + Vehicle structured data + hreflang
- Privacy Policy and Terms of Service (EN + Bangla)
- Phase 1 recommendation (rule-based: location + search budget)

V2.0 — Month 3–6

- BanglaWheel Bid — live auction with Socket.io
- Credibility Report and Garage Portal
- Warranty escrow and claim flow
- EMI Calculator
- Buyer Education Hub with Bangla article support
- BRTA guide and fee calculator
- Saved search + SMS alerts
- Phase 2 recommendation (behavioural tracking + affordability midpoint)

V3.0 — Month 6–12

- Financing lead forms (bank integrations)
- Car comparison (up to 3 cars)
- PWA — installable on Android
- OBD-II scan upload by garage
- Seller performance ratings
- BRTA API chassis verification
- Phase 3 recommendation — collaborative filtering

19. Reference Platforms

Study each platform before writing code.

Platform	URL	What to Study
Carro	carro.sg	Trust signal placement. Certified badge UI.
Carsome	carsome.my	Best mobile UX. Inspection display. Filter UX.
Cars24	cars24.com	Car comparison. Price estimator. Sell-from-home.
AutoTrader CA	autotrader.ca	Dealer dashboard. Analytics. Saved search alerts.
PakWheels	pakwheels.com	Most relevant — same JDM import market. Study everything.
CarGurus	cargurus.com	Price badge: Great Deal / Fair Deal.
Hagerty	hagerty.com	Best auction UX — live bid feed, countdown, auto-bid.
BidExport	bidexport.com	Auction mechanics — reserve handling, anti-sniping.